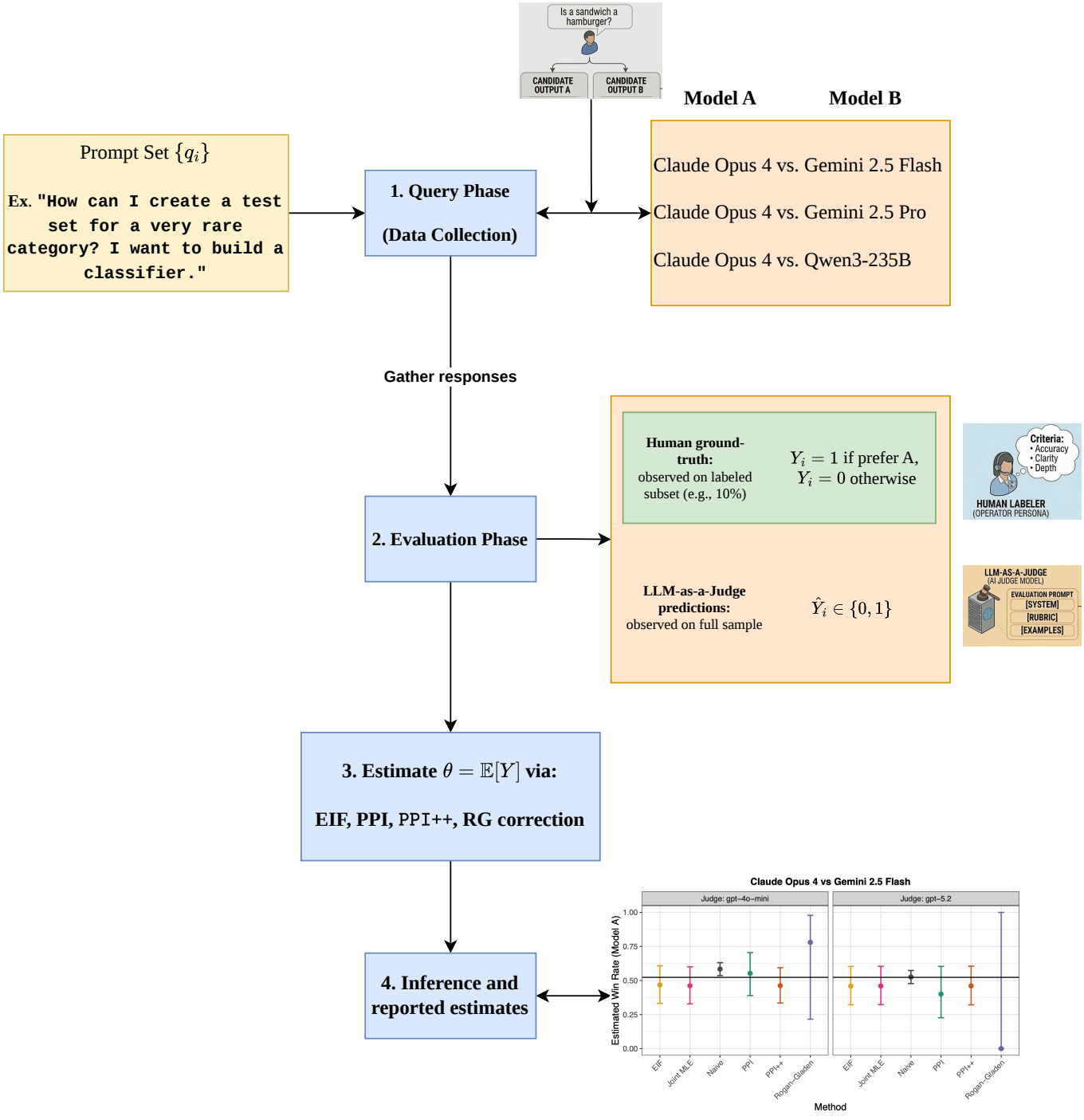
Is a sandwich a hamburger?
CANDIDATE OUTPUT A
CANDIDATE OUTPUT B
Model A
Model B
Prompt Set $\{q_i\}$
Ex. "How can I create a test set for a very rare category? I want to build a classifier."
1. Query Phase
(Data Collection)
Claude Opus 4 vs. Gemini 2.5 Flash
Claude Opus 4 vs. Gemini 2.5 Pro
Claude Opus 4 vs. Qwen3-235B
Gather responses
2. Evaluation Phase
Human ground-truth:
observed on labeled subset (e.g., 10%)
$Y_i = 1$ if prefer A,
$Y_i = 0$ otherwise
LLM-as-a-Judge predictions:
observed on full sample
$\hat{Y}_i \in \{0, 1\}$
Criteria:
• Accuracy
• Clarity
• Depth
HUMAN LABELER
(OPERATOR PERSONA)
LLM-AS-A-JUDGE
(AI JUDGE MODEL)
EVALUATION PROMPT
[SYSTEM]
[RUBRIC]
[EXAMPLES]
3. Estimate $\theta = \mathbb{E}[Y]$ via:
EIF, PPI, PPI++, RG correction
4. Inference and reported estimates
Claude Opus 4 vs Gemini 2.5 Flash
Judge: gpt–4o–mini
Judge: gpt–5.2
Estimated Win Rate (Model A)
1.00
0.75
0.50
0.25
0.00
EIF
Joint MLE
Naive
PPI
PPI++
Rogan–Gladen
Method